

ACCTG 211H — Mechanics Drill Set (Answer Key)

Week 4 — Information and Incentives

Week 4 drills: information and incentives

Complete after Day 8 and submit Sunday, February 7. These topics are not on Quiz 1. Work individually, check the key, and correct errors. Graded for completion; no generative AI.

1. Among 100 cases, 25 are high demand and 75 low demand. A report is favorable in 80% of high cases and 20% of low cases. Complete a two-by-two frequency table. What fraction of favorable reports correspond to high demand? Why is the answer not 80%?
2. Return to the chapter's 50% prior and the same report accuracy. After a favorable report, the probability of high demand is 80%. Payoffs one year later are \$880 or \$440. At 10%, calculate expected payoff and value after each report.
3. At that report date an investor can buy the claim for \$700 or decline. Compute the report's expected decision benefit before its fee. Should the investor purchase information costing \$8? What about \$11? All costs and values are at that decision date.
4. For the main text's manager, compare fixed pay of \$40 with \$20 plus \$80 for a good outcome. Calculate the manager's net expected payoff under each effort choice and identify the choice. Extra effort costs \$30 and raises the good-outcome probability from 40% to 80%.
5. Classify each: recognizing interest under a known contract; estimating a scooter's useful life; offering a warranty that costs more for unreliable products; omitting interest because a bill has not arrived. Which can involve private information, and which does not require it?

Self-Check Answers

1. High: favorable **20**, unfavorable **5**. Low: favorable **15**, unfavorable **60**. Posterior high given favorable = $20/35 = 57.14\%$. The prior is only 25%; a test's detection rate is not generally the posterior probability.
2. Favorable: expected payoff **\$792**, value **\$720**. Unfavorable: expected payoff **\$528**, value **\$480**.
3. Without the report, decline (\$600 value < \$700 price). With it, buy only after favorable news, giving expected benefit $0.5 \times (720 - 700) = \mathbf{\$10}$. An \$8 fee leaves **\$2**; an \$11 fee leaves **-\$1**, so decline that information offer.
4. Fixed pay: ordinary **\$40**, extra **\$10**; choose ordinary. Bonus: ordinary **\$52**, extra **\$54**; choose extra. Both chosen payoffs meet the \$40 outside option. These are two feasible contracts, not proof of global optimality.
5. Respectively: **routine recognition; estimation; potentially costly signaling; misstatement** under the stated facts. Useful-life estimates may incorporate private knowledge; a warranty can reveal type under suitable incentives. Fixed contractual interest recognition need not involve private information.